SPHERE MISSION CONTROL

THERMAL REVISION DECK

PART 1: CORE THERMODYNAMICS & COOLING EQUATIONS

STEP 1: Cut along outer dashed lines. **STEP 2:** Fold vertically along center blue dotted line. **STEP 3:** Punch top-left ring hole.

CUT BORDER

CARD #01 — THERMODYNAMICS

NEWTON'S LAW OF COOLING

Question: What fundamental equation governs the rate of heat loss of a hot core capsule submerged in a cooler ambient fluid?

FOLD

FRONT: QUESTION — SPHERE STUDY DECK

CARD #01 — KEY FORMULA

EXPONENTIAL DECAY

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$$

Explanation: The cooling rate $\frac{dT}{dt}$ is directly proportional to the instantaneous temperature gap $(T - T_{\text{env}})$.

BACK: REVEAL & LAW — k = cooling constant

CUT BORDER

CARD #02 — THERMODYNAMICS

DECAY CONSTANT (k)

Question: What does the constant k quantify physically, and which capsule is better insulated: $k = 0.15\ \text{min}^{-1}$ or $k = 0.015\ \text{min}^{-1}$?

FOLD

FRONT: QUESTION — SPHERE STUDY DECK

CARD #02 — INSULATION FACTOR

CAPSULE B (k = 0.015)

Answer: k measures heat loss speed ($k = \frac{hA}{mc}$). A **smaller k value** means slower cooling and superior thermal insulation (heat held 10x longer!).

BACK: REVEAL & LAW — Units: min^{-1} or s^{-1}

CUT BORDER

CARD #03 — HEAT TRANSFER

FOURIER'S CONDUCTION LAW

Question: How does thermal energy transfer directly through solid matter, and what formula defines conduction heat flow?

FOLD

FRONT: QUESTION — SPHERE STUDY DECK

CARD #03 — CONDUCTION EQUATION

MOLECULAR VIBRATION

$$q = -k_{\text{mat}} \cdot A \cdot \frac{dT}{dx}$$

Explanation: Kinetic energy spreads by vibrating atoms bumping adjacent particles. Low-k_{mat} materials (cotton, fiberglass) act as thermal insulators.

BACK: REVEAL & LAW — k_mat = conductivity

CUT BORDER

CARD #04 — EQUILIBRIUM

THERMAL EQUILIBRIUM

Question: When does net heat transfer between the capsule core and the 0°C ice bath drop to zero? What physical law governs this?

FOLD

FRONT: QUESTION — SPHERE STUDY DECK

CARD #04 — ZEROTH LAW

EQUAL TEMPERATURES

Answer: When $T_{\text{capsule}} = T_{\text{env}} = 0^\circ\text{C}$. At this point, temperature gradient $\Delta T = 0$, so $\frac{dQ}{dt} = 0$. Governed by the **Zeroth Law of Thermodynamics**.

BACK: REVEAL & LAW — Net ΔQ = 0 Joules

THERMAL REVISION DECK

PART 2: RADIATIVE SHIELDING & MULTI-LAYER INSULATION (MLI)

STEP 1: Cut along outer dashed lines. **STEP 2:** Fold vertically along center blue dotted line. **STEP 3:** Punch top-left ring hole.

CUT BORDER

CARD #05 RADIATION

SPACE VACUUM RADIATION

Question: Why is thermal radiation the ONLY heat transfer mode that can operate across the vacuum of deep space?

FOLD

FRONT: QUESTION SPHERE STUDY DECK

CARD #05 STEFAN-BOLTZMANN

PHOTON HEAT FLOW

$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_{\text{env}}^4)$$

Answer: Vacuum contains zero molecules for conduction or convection. Heat radiates via infrared electromagnetic waves carrying energy without matter!

BACK: REVEAL & LAW $\sigma = 5.67\times10^{-8}$ W/m²K⁴

CUT BORDER

CARD #06 SHIELDING

MYLAR RADIATIVE SHIELD

Question: How does a thin shiny aluminized Mylar foil protect spacecraft and astronauts from extreme solar thermal radiation?

FOLD

FRONT: QUESTION SPHERE STUDY DECK

CARD #06 OPTICAL REFLECTION

INFRARED REFLECTANCE

Answer: Mylar's vapor-deposited aluminum coating has high reflectivity ($> 90\%$) and very low emissivity ($\epsilon < 0.05$), bouncing infrared rays back like an optical mirror.

BACK: REVEAL & LAW Reflects >90% IR heat

CUT BORDER

CARD #07 CONVECTION

CONVECTIVE AIR TRAPPING

Question: Why is bubble wrap such an effective insulator during atmospheric lab trials? What mechanism does it eliminate?

FOLD

FRONT: QUESTION SPHERE STUDY DECK

CARD #07 STAGNANT CELLS

STAGNANT AIR POCKETS

Answer: Air is a poor conductor ($k_{\text{air}} \approx 0.026\ \text{W/mK}$). Sealed bubble cells prevent **buoyancy circulation currents** from moving heat away!

BACK: REVEAL & LAW Rayleigh convection stopped

CUT BORDER

CARD #08 MLI SPACESUITS

MULTI-LAYER INSULATION

Question: Why do spacesuits and satellites stack Cotton + Bubble Wrap + Mylar in composite layers?

FOLD

FRONT: QUESTION SPHERE STUDY DECK

CARD #08 TRIPLE SHIELD

TRIPLE HEAT SHIELD

Answer: Composite synergy:
- **Cotton:** Blocks solid conduction.
- **Bubble Wrap:** Traps fluid convection.
- **Mylar:** Reflects infrared radiation.

BACK: REVEAL & LAW All 3 heat paths blocked

THERMAL REVISION DECK

PART 3: DIGITAL TWINS, TELEMETRY & SPACE ANALOGS

STEP 1: Cut along outer dashed lines. **STEP 2:** Fold vertically along center blue dotted line. **STEP 3:** Punch top-left ring hole.

CUT BORDER

CARD #09 — SIMULATION

DIGITAL TWIN CONCEPT

Question: What is a Digital Twin in modern aerospace engineering, and how is it used in SPHERE Mission Control?

FOLD

FRONT: QUESTION — SPHERE STUDY DECK

CARD #09 — VIRTUAL TELEMETRY

LIVE VIRTUAL MODEL

Answer: A high-fidelity physics model running equations in real-time alongside live physical sensors, enabling immediate anomaly detection and residual tracking.

BACK: REVEAL & LAW — Hardware + Simulation

CUT BORDER

CARD #10 — LAB ANALOG

ICE BATH SPACE ANALOG

Question: Why do we submerge the insulated capsule into a $0°\mathrm{C}$ ice-water slurry during laboratory mission runs?

FOLD

FRONT: QUESTION — SPHERE STUDY DECK

CARD #10 — CONSTANT SINK

CONSTANT HEAT SINK

Answer: Melting ice water stays at an exact, invariant $0°\mathrm{C}$ ($T_{\mathrm{env}} = 0°\mathrm{C}$), simulating the immense, infinite thermal heat sink of deep space.

BACK: REVEAL & LAW — $T_{\mathrm{env}} = 0.0°\mathrm{C}$ constant

CUT BORDER

CARD #11 — THERMAL MASS

HEAT CAPACITY OF WATER

Question: Why is liquid water ($c = 4184\ \mathrm{J/kg} \cdot {}^\circ\mathrm{C}$) ideal for capsule core telemetry trials? What is the heat formula?

FOLD

FRONT: QUESTION — SPHERE STUDY DECK

CARD #11 — THERMAL ENERGY

HIGH THERMAL MASS

$$Q = m \cdot c \cdot \Delta T$$

Answer: Water stores large thermal energy per gram, producing smooth, measurable exponential decay over $15+$ minutes without dropping instantly.

BACK: REVEAL & LAW — $c = 4184\ \mathrm{J/kg} \cdot \mathrm{K}$

CUT BORDER

CARD #12 — GRAPH ANALYSIS

DECAY CURVE CURVATURE

Question: Why is the temperature-time graph $T(t)$ steep at $t = 0$ and increasingly flat toward $t = 15\ \mathrm{min}$?

FOLD

FRONT: QUESTION — SPHERE STUDY DECK

CARD #12 — ASYMPTOTIC FLOW

SHRINKING TEMP GAP

Answer: Heat flux rate $\frac{dQ}{dt} \propto (T - T_{\mathrm{env}})$. As the capsule cools, $(T - T_{\mathrm{env}})$ shrinks, causing the rate of temperature drop to slow down asymptotically.

BACK: REVEAL & LAW — $\mathrm{Slope} = -k(T - T_{\mathrm{env}})$

THERMAL REVISION DECK

PART 4: MISSION PROTOCOLS, MATH METRICS & SYNTHESIS

STEP 1: Cut along outer dashed lines. **STEP 2:** Fold vertically along center blue dotted line. **STEP 3:** Punch top-left ring hole.

CUT BORDER

CARD #13 METRICS

CORRELATION SCORE (R²)

Question: How does Mission Control calculate the Correlation Score between live physical sensor telemetry and the Digital Twin model?

FOLD

FRONT: QUESTION SPHERE STUDY DECK

CARD #13 MODEL ACCURACY

RESIDUAL VARIANCE

$$R^2 = 1 - \frac{\sum (T_{\text{meas}} - T_{\text{pred}})^2}{\sum (T_{\text{meas}} - \bar{T})^2}$$

Answer: Quantifies goodness-of-fit. A score $> 95\%$ proves the theoretical physics model matches actual mission hardware telemetry with high fidelity.

BACK: REVEAL & LAW Target: >95% Correlation

CUT BORDER

CARD #14 LAB PROTOCOL

PROBE PLACEMENT

Question: Why must the digital thermocouple probe be centered inside the fluid rather than touching the capsule wall?

FOLD

FRONT: QUESTION SPHERE STUDY DECK

CARD #14 BOUNDARY LAYER

BULK FLUID TEMPERATURE

Answer: Touching the outer wall measures the cold boundary gradient rather than core liquid bulk temperature, introducing measurement artifacts into telemetry.

BACK: REVEAL & LAW Center probe exactly

CUT BORDER

CARD #15 SI UNITS

SI UNITS REFERENCE

Question: What are the standard SI units for temperature T, heat flow Q, thermal conductivity k_{mat}, and cooling constant k?

FOLD

FRONT: QUESTION SPHERE STUDY DECK

CARD #15 STANDARD UNITS

UNITS CHEAT SHEET

- **Temperature (T):** Kelvin (K) or Celsius $(^\circ\text{C})$
- **Thermal Energy (Q):** Joules (J)
- **Decay Constant (k):** s^{-1} or min^{-1}
- **Conductivity (k_{mat}):** $\text{W}/(\text{m}\cdot\text{K})$

BACK: REVEAL & LAW Standard Metric Units

CUT BORDER

CARD #16 CREW STRATEGY

CREW MLI OPTIMIZATION

Question: Which insulation configuration produces the highest survival time and smallest decay constant k? Why?

FOLD

FRONT: QUESTION SPHERE STUDY DECK

CARD #16 SURVIVAL RECORD

3-LAYER COMPOSITE MLI

Answer: Cotton + Bubble Wrap + Mylar drops decay constant k from $\sim 0.150\ \text{min}^{-1}$ (bare tube) down to $< 0.020\ \text{min}^{-1}$, preserving capsule heat for $>$ 20 minutes.

BACK: REVEAL & LAW Maximum Thermal Shield